



CSE461

Introduction to Robotics Lab

Lab No. : 01

Group : 01

Section : 06

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Submitted to - CSDM

1. Objectives:

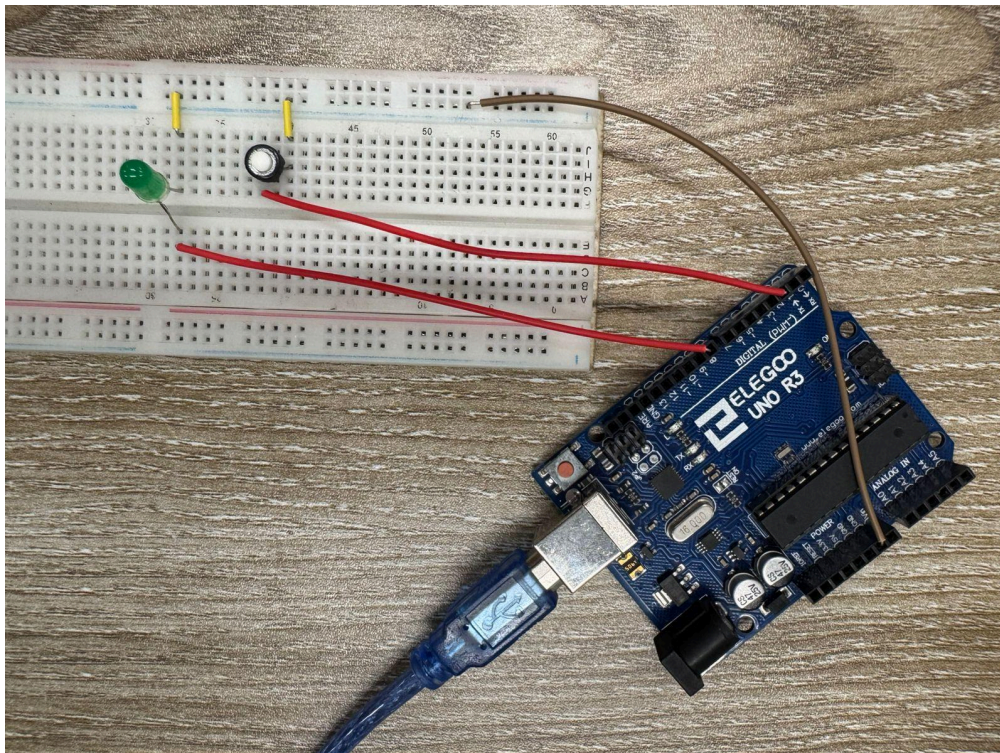
This task's objective focuses on a combination of two operations: experiment 1's LED blinking and experiment 2's reading switch input - to control a LED using a switch. This will result in : when the LED will TURN ON when the switch is pressed and when switch is released or not pressed, LED will TURN OFF.

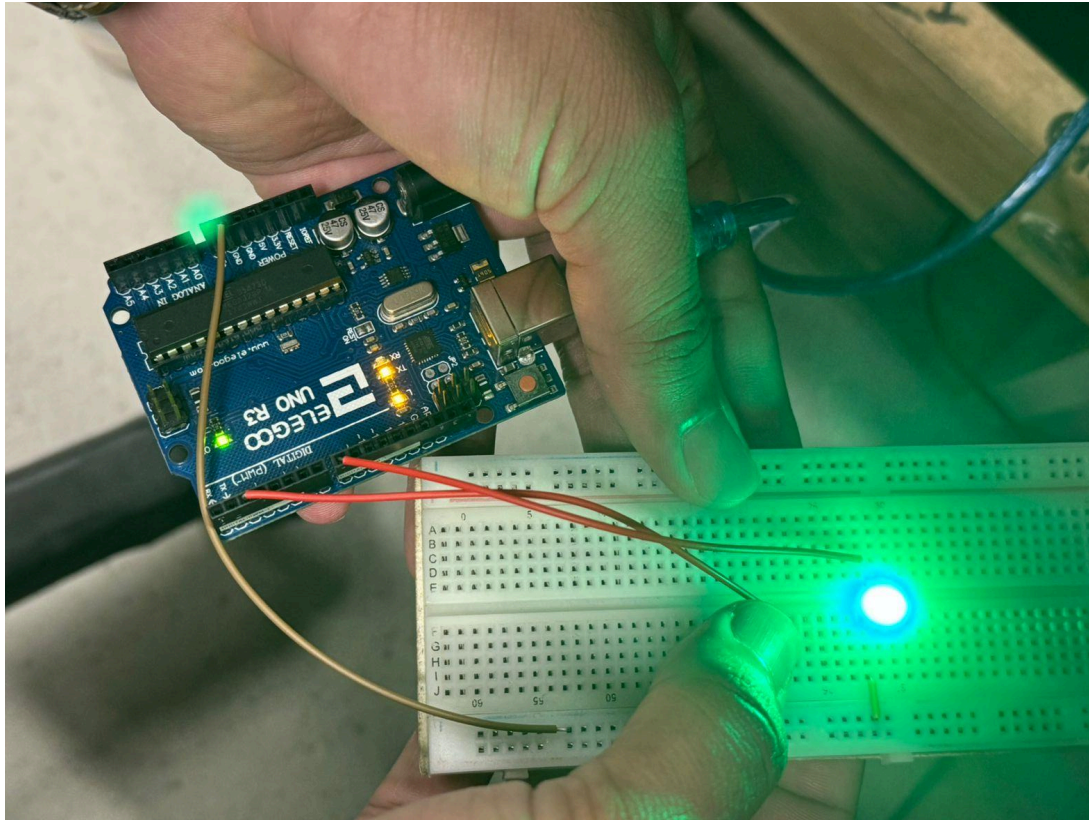
2. Equipments:

- Arduino Uno R3
- LED (Light Emitting Diode)
- Push-button switch
- Resistor (220Ω for LED)
- Breadboard
- Jumper wires

3. Experimental Setup:

(Picture + Explanation)





Here, we connected one terminal of the switch to the GND and the other terminal of the switch to digital pin 2 on the Arduino ; We connected the LED to digital pin 8 on the Arduino and the other terminal of the LED to the GND. Lastly, We shorted all the GND's with the GND pin on the Arduino.

4. Code: (If Applicable)

```
#define BUTTON_PIN 2 // Define the button pin
#define LED_PIN 8
void setup() {
  pinMode(BUTTON_PIN, INPUT_PULLUP); // Set the button pin as input
  Serial.begin(9600); // Initialize serial communication
  pinMode(LED_PIN, OUTPUT); // Set the LED pin as an output
}
void loop() {
  if (digitalRead(BUTTON_PIN) == LOW) { // Button is pressed
    Serial.println("Button pressed."); // Print message to S. Monitor
    digitalWrite(LED_PIN, HIGH); // Turn the LED on
    Serial.println("LED ON"); // Print LED status
  }else{
```

```
Serial.println("Button not pressed.");  
digitalWrite(LED_PIN, LOW); // Turn the LED off  
}  
delay(100); // Small delay to debounce the button  
}
```

5. Results (Output of the experiment):

When the switch was pressed, the LED turned ON and it showed in the terminal of Arduino as “Button Pressed.” and when the switch was not pressed, the LED turned off and it showed in the terminal of Arduino as “Button not pressed.” So, when the switch was not pressed, the pin read HIGH and when pressed, it connected to GND - resulting in a LOW reading.

6. Discussions :

From this task, we got to learn how to control an LED with a push-button switch using an Arduino Uno R3 microcontroller. For this task, we had to use what we learnt from the previous tasks : combining the concepts from Experiment 1 (LED blinking) and Experiment 2 (reading switch input) to achieve LED control.

No significant problems were encountered during this experiment, the setup and code performed as expected, allowing for real-time control of the LED based on the switch's state. Thus, this lab provided a foundational understanding of configuring digital input and output pins, using internal pull-up resistors - which are crucial skills in robotics.